FIT3155: Week 12 Lab Sheet

(Questions are based on the Week 11 topic of Minimum-Cost Bipartite Graph Matching.)

Objectives: To develop an understanding of bipartite graph matching and how the Hungarian Algorithm solves the minimum-cost bipartite matching problem and its link to linear programming and network flow problems.

Conceptual exercises

☐ 1. Given an $N \times N$ cost matrix $C = [c_{ij}]$ and its associated bipartite graph/clique, formulate the minimum-cost bipartite matching problem using binary decision variables $X = [x_{ij}]$ as a mathematical optimization problem.

☐ 2. Let $\{u_i\}$ and $\{v_j\}$ be dual variables associated with the rows and columns of $C = [c_{ij}]$.

☐ (a) Show that the objective function of the linear assignment problem can be transformed into the following form:

$$z = \sum_{i=1}^{N}\sum_{j=1}^{N}(c_{ij} - u_i - v_j)x_{ij} + \sum_{i=1}^{N} u_i + \sum_{j=1}^{N} v_j.$$

☐ (b) Assume that $c_{ij} = u_i + v_j$ whenever $x_{ij} = 1$. Reason why under this equality condition the objective simplifies to

$$z = \sum_{i=1}^{N} u_i + \sum_{j=1}^{N} v_j.$$

☐ (c) Assume that $c_{ij} \geq u_i + v_j$ whenever $x_{ij} = 0$ and $c_{ij} = u_i + v_j$ whenever $x_{ij} = 1$. Show that the minimum-cost bipartite matching problem can be equivalently modelled as

$$\max \quad \sum_{i=1}^{N} u_i + \sum_{j=1}^{N} v_j$$

subject to

$$u_i + v_j \leq c_{ij} \qquad \forall i, j.$$

☐ 3. A company has four trucks located at different garages. Each truck must be assigned to exactly one of four different sites, with no site receiving more than one truck. The distances (in kilometers) between the truck locations and the sites are given below.

	Site A	Site B	Site C	Site D
Truck 1	90	75	75	80
Truck 2	35	85	55	65
Truck 3	125	95	90	105
Truck 4	45	110	95	115

Assign each truck to a site so as to minimize the total distance traveled.

4. A job consultant has four clients (each advertising a job) and five applicants looking to be hired. The consultant is able to reliably rank the suitability of each applicant to the clients' job description on a scale of zero (poorest fit) to ten (best fit). The consultant's rankings are given in the following table.

	Appl. 1	Appl. 2	Appl. 3	Appl. 4	Appl. 5
Client 1	7	4	7	3	10
Client 2	5	9	3	8	7
Client 3	3	5	6	2	9
Client 4	6	5	0	4	8

The consultant wants to match the applicants to clients and in doing so aims to **maximize** (not minimize) the sum of the matched rankings. Can this be solved using the Hungarian approach? If so, how is the above problem converted to an insance of min-cost bipartite graph matching to be able to do so?

The following questions are intended to highlight connections with the Network Flows topic covered in FIT2004. FIT3155 will not assess knowledge of network flows and its applications in the final exam, as this is assumed prior knowledge. However, as part of your learning it remains useful to understand these connections.

5. The above questions were instances of minimum-cost bipartite matching problems. A related variant is the maximum **cardinality** bipartite matching problem, whose objective is to match as many vertices as possible in a bipartite graph such that each vertex is incident to at most one matched edge in the resultant matching. How can a network flow formulation be adapted to solve this problem?

6. Step 3 of the Hungarian algorithm discussed in the lecture slides involves assigning a set of 0-cells and then determining the minimum number of lines required to cover all zeros in the current state of the (reduced cost) matrix. Can a network flow algorithm be employed to carry out Step 3 as an alternative? Justify your answer.

7. (This question concerns modelling the following as a network flow problem.) The coordinator of Monash Faculty of IT's IBL program has a cohort of n students wanting to be placed with $m < n$ companies. Each company $1 \leq i \leq m$ has at most x_i available places for Monash students. Each student nominates exactly k companies (in no particular order) that they wish to be placed with.

After interviews, each company nominates $y_i > x_i$ students (again in no particular order) whom they would be happy to host, subject to their capacity constraint x_i.

Taking the respective student–company nominations into account, the IBL coordinator's goal is to assign the maximum number of students to mutually agreeable companies. Can you help the IBL coordinator address this problem?

Implementation exercise

(A message of caution: 'Vibe coding' with GenAI will hinder your learning of algorithms and data structures. The degree (and this unit) you have enrolled into is a specialist degree (unit). By definition it expects one to demonstrate specialist understanding of topics. This understanding comes from making a concerted effort, via thinking, implementing, debugging etc.)

☐ 1. Write a program to find minimum-cost matching of any input weighted bipartite graph. For inputs, use above instances (also consider generating random instances).

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